SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Dry Run Evaluation

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 2 – Dry Run Evaluation
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	D/ 2026	**Date:**	05/10/2026

Code of Conduct

I G.Sanghamitra (1924260351) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.Sa

Instructions

- Read each question carefully; answer all four questions (25 marks each, 100 in total).
- Show every dry-run step: the initial queue, stack, priority queue or belief state, each iteration, your assumptions and a clear final answer, and finish with the interpretation asked for.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Dry Run Evaluation rubric: understanding of the search problem, state expansion and dry run execution, application of the search algorithm, accuracy of the solution path, and presentation and logical reasoning.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end). Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q2**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**

7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**
9. Uninformed Search Strategies	Dry runs of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Question 1: Agents, Environments and Rationality – Dry Run of Agent Programs

Topic: Intelligent agents, rationality, nature of environments, structure of agents | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives [its location, status of its own square] and performs one action: Suck, Right, Left or NoOp. The agent starts in square A, both squares are initially Dirty, and dirt never reappears.

Performance measure over T = 6 time steps: +10 for every square cleaned (a Suck on a Dirty square) and −1 for every Left or Right move.

Agent X (simple reflex): if the current square is Dirty, Suck; otherwise move to the other square.

Agent Y (model-based reflex): keeps an internal state of the squares it knows to be clean. If the current square is Dirty, Suck; otherwise, if both squares are known clean, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent. **[4 Marks]**

(b) Classify the task environment on observability, number of agents, determinism, episodic/sequential, static/dynamic and discrete/continuous, with a one-phrase justification each. **[4 Marks]**

(c) Dry-run both agents for all 6 time steps. Complete the table (percept, action, and Y's internal state after the step). **[9 Marks]**

Step	X: Percept	X: Action	Y: Percept	Y: Action	Y: Known-Clean Squares
1					
2					
3					
4					
5					
6					

(d) Compute the total performance of X and of Y over the 6 steps. **[3 Marks]**

(e) Interpret: which agent is rational? Both agents receive the identical percept [B, Clean] at step 4 – explain why they act differently, using the structure of the two agent programs, and distinguish a rational agent from an omniscient agent. **[5 Marks]**

Question 2: Foundations and History of AI; Problem-Solving Agent and Problem Formulation

Topic: Foundations and history of AI, problem-solving agents, problem formulation | 15 minutes

Part A. Complete the table by classifying each system or idea under one of the four approaches to AI (thinking humanly, acting humanly, thinking rationally, acting rationally) and naming the era or milestone of AI history to which it belongs.

(a) Complete the table. **[8 Marks]**

System / Idea	Approach to AI	Era / Milestone
Turing Test proposed (1950)		
General Problem Solver (Newell and Simon, 1957)		
Aristotle's syllogisms and formal logic		
An agent that does the right thing to maximise expected performance		

Part B. A problem-solving agent has two jugs: X of capacity 4 litres and Y of capacity 3 litres, both initially empty and without measuring marks. It may fill a jug from a tap, empty a jug on the ground, or pour one jug into the other until the source is empty or the target is full. The goal is to have exactly 2 litres in jug X.

(b) Formulate the problem: state representation, initial state, actions, transition model, goal test and path cost. **[6 Marks]**

(c) Generate all successor states of (0, 0) and of (4, 0) (leave out actions that leave the state unchanged). **[3 Marks]**

(d) How many states does the state space contain? Write a shortest solution as a sequence of states and give its path cost. **[3 Marks]**

(e) Interpret: describe the four phases (goal formulation, problem formulation, search, execution) of a problem-solving agent for this task, and explain why the agent can execute the found solution without looking at new percepts. **[5 Marks]**

Question 3: Searching for Solutions – Uninformed Search Strategies

Topic: Searching for solutions, BFS, DFS, uniform-cost search | 15 minutes

Consider the weighted undirected graph below. The start node is S and the goal node is G. Use graph search with an explored set; expand neighbours alphabetically, test the goal when a node leaves the frontier, and break ties alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them.

Edge	Cost	Edge	Cost
S – A	1	B – D	1

S – B	4	C – G	6
A – C	2	D – G	2
A – D	5		

(a) Perform BFS from S (queue initially [S]). Do not add a node that is already explored or already in the queue. **[6 Marks]**

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1			
2			
3			
4			
5			
6			

(b) Perform DFS from S using a stack (initially [S]); push children so that the alphabetically first child is popped first. Write the stack with its top on the left. **[5 Marks]**

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		

(c) Perform uniform-cost search from S. Write queue entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found. **[7 Marks]**

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

(d) Complete the comparison table. **[4 Marks]**

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(e) Interpret: explain why UCS removed as many nodes as BFS yet found a cheaper path, why DFS needed the fewest nodes here but cannot be trusted in general, and why an explored set is needed in graph search (use iteration 3 of BFS as an example). **[3 Marks]**

Question 4: Search with Partial Information – Belief States

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search) | 15 minutes

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

(a) Initially the agent has no sensors and does not know its location or the dirt status. Formulate the problem over belief states: initial belief state, actions, goal test and path cost. **[3 Marks]**

(b) Apply the action sequence [Right, Suck, Left, Suck] and write the belief state after each action. **[6 Marks]**

Step	Action	Belief State after Action
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}
1	Right	
2	Suck	
3	Left	
4	Suck	

(c) Now the agent has a local sensor that reports [location, status of the current square]. Starting again from the initial belief state, the first percept is [Left, Dirty]; the agent then performs Suck and perceives [Left, Clean]; it then performs Right and perceives [Right, Dirty]. Fill in the belief state after each action (prediction) and after each percept (update). **[8 Marks]**

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(d) How many different belief states are possible when there are 8 physical states? Why does this make the problem harder than searching over physical states? **[3 Marks]**

(e) Interpret: explain why the percepts in part (c) shrink the belief state, how this differs from the sensorless case in part (b), and why the sensorless plan can still be called rational. **[5 Marks]**

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Accuracy of Solution Path	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Q. No. / Criteria	Understanding of Search Problem	State Expansion and Dry Run Execution	Application of Search Algorithm	Accuracy of Solution Path	Accuracy of Solution Path	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SOLUTIONS

Question 1: Agents, Environments and Rationality

(a) PEAS Description

Component	Description
Performance Measure	+10 points for each square cleaned and −1 point for every Left or Right movement.
Environment	Two squares, A and B, containing dirt. Dirt does not reappear after cleaning.
Actuators	Suck, Left, Right, and NoOp.
Sensors	Detect the agent's current location and the dirt status of the current square.

(b) Classification of the Task Environment

Property	Classification	Justification
Observability	Partially observable	The agent observes only its current location and the status of its current square.
Number of agents	Single-agent	Only one vacuum-cleaner agent operates in the environment.
Determinism	Deterministic	Each action produces a predictable result.
Episodic / Sequential	Sequential	Each action affects the future state and subsequent decisions.
Static / Dynamic	Static	Dirt does not reappear, and the environment does not change independently during an action.
Discrete / Continuous	Discrete	Locations, actions, and dirt conditions have finite possible values.

(c) Dry Run of Agent X and Agent Y

Initial conditions: both agents start in A; both squares are dirty. Agent X uses only the current percept. Agent Y records the squares known to be clean.

Step	X: Percept	X: Action	Y: Percept	Y: Action	Y's known-clean squares
1	[A, Dirty]	Suck	[A, Dirty]	Suck	{A}
2	[A, Clean]	Right	[A, Clean]	Right	{A}
3	[B, Dirty]	Suck	[B, Dirty]	Suck	{A, B}
4	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}
5	[A, Clean]	Right	[B, Clean]	NoOp	{A, B}
6	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}

(d) Total Performance

Performance = 10 × successful cleaning actions − number of movements.

Agent X: 2 successful cleaning actions and 4 movements. Score = (2 × 10) − 4 = **16**.

Agent Y: 2 successful cleaning actions and 1 movement. Score = (2 × 10) − 1 = **19**.

(e) Interpretation

Agent Y is more rational for the stated performance measure because it remembers which squares are clean and avoids unnecessary movement. At Step 4, both agents receive [B, Clean]. Agent X uses a simple reflex rule and moves Left because the current square is clean. Agent Y has an internal state showing that both A and B are clean, so it chooses NoOp. A rational agent chooses actions expected to maximize performance based on available information.
An omniscient agent would know the actual state and outcomes in advance; rationality does not require omniscience.

Question 2: Foundations and History of AI; Problem-Solving Agent

Part A – (a) Four Approaches to AI

System / Idea	Approach to AI	Era / Milestone
Turing Test proposed (1950)	Acting humanly	Early AI era; a test of machine intelligence proposed by Alan Turing.
General Problem Solver (Newell and Simon, 1957)	Thinking humanly	Early symbolic AI; an attempt to model human problem-solving.

Aristotle's syllogisms and formal logic	Thinking rationally	Ancient foundations of logic and rule-based reasoning.
Agent that maximizes expected performance	Acting rationally	Rational-agent approach; choosing actions to maximize expected performance.

(b) Water Jug Problem Formulation

Component	Description
State representation	(x, y), where x is the amount in 4-L jug X and y is the amount in 3-L jug Y.
Initial state	(0, 0)
Actions	Fill X, Fill Y, Empty X, Empty Y, Pour X into Y, Pour Y into X.
Transition model	Actions change quantities according to capacities; pouring continues until the source is empty or the target is full.
Goal test	x = 2; jug X contains exactly 2 litres.
Path cost	Each action costs 1; total cost is the number of actions.

Valid state ranges: $0 \le x \le 4$ and $0 \le y \le 3$.

(c) Successor States

From (0, 0): Fill X → (4, 0); Fill Y → (0, 3). Other actions leave the state unchanged.

From (4, 0): Empty X → (0, 0); Fill Y → (4, 3); Pour X into Y → (1, 3).

(d) State Space and Shortest Solution

The number of possible states is $(4 + 1)(3 + 1) =$ **20 states**.

Step	Action	State (X, Y)
0	Initial state	(0, 0)
1	Fill Y	(0, 3)
2	Pour Y into X	(3, 0)
3	Fill Y	(3, 3)
4	Pour Y into X until X is full	(4, 2)
5	Empty X	(0, 2)
6	Pour Y into X	(2, 0)

The goal is reached with exactly 2 litres in X. Path cost = **6 actions**.

(e) Four Phases of a Problem-Solving Agent

- **Goal formulation:** Define the goal of obtaining exactly 2 litres in jug X.
- **Problem formulation:** Specify states, initial state, actions, transition model, goal test, and path cost.
- **Search:** Find a sequence of actions from (0, 0) to a state where x = 2.
- **Execution:** Carry out the selected sequence of actions.

The agent can execute the plan without new percepts because the initial state and action outcomes are assumed known and the environment is deterministic. The plan works provided the actions behave as specified.

Question 3: Searching for Solutions – Uninformed Search

The graph is undirected. Edge costs are S–A = 1, S–B = 4, A–C = 2, A–D = 5, B–D = 1, C–G = 6, and D–G = 2.
Neighbours are considered alphabetically, and the goal is tested when removed from the frontier.

(a) Breadth-First Search (BFS)

Initial queue: [S]. BFS uses a FIFO queue and ignores edge costs during exploration.

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[A, B]	{S}
2	A	[B, C, D]	{S, A}
3	B	[C, D]	{S, A, B}
4	C	[D, G]	{S, A, B, C}
5	D	[G]	{S, A, B, C, D}
6	G	[]	{S, A, B, C, D, G}

BFS path: S → A → C → G. Path cost = 1 + 2 + 6 = **9**.

(b) Depth-First Search (DFS)

Initial stack: [S]. The top of the stack is on the left. Push children in reverse alphabetical order so the alphabetically first child is popped first.

Iteration	Node popped	Stack after pushing children
1	S	[A, B]
2	A	[C, D, B]
3	C	[G, D, B]
4	G	[D, B]

DFS path: S → A → C → G. Path cost = 1 + 2 + 6 = **9**.

(c) Uniform-Cost Search (UCS)

Initial priority queue: [S:0]. Entries use Node:g(parent); lower cumulative cost is expanded first.

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	[A:1(S), B:4(S)]

2	A	1	[C:3(A), B:4(S), D:6(A)]
3	C	3	[B:4(S), D:6(A), G:9(C)]
4	B	4	[D:5(B), G:9(C)]
5	D	5	[G:7(D)]
6	G	7	[]

When B is expanded, D improves from cost 6 to 5. When D is expanded, G improves from cost 9 to 7. UCS path: S → B → D → G. Path cost = 4 + 1 + 2 = **7**.

(d) Comparison

Strategy	Path found	Path cost	Nodes removed	Optimal here?
BFS	S → A → C → G	9	6	No
DFS	S → A → C → G	9	4	No
UCS	S → B → D → G	7	6	Yes

(e) Interpretation

BFS and UCS each remove six nodes in this graph, but BFS chooses by depth while UCS chooses by cumulative path cost. Therefore, UCS finds the cheaper path (cost 7 instead of 9). DFS reaches G after four removals here, but its path is not least-cost and DFS is not generally optimal. An explored set avoids repeated expansion; for example, at BFS iteration 3, D is already in the queue when B is expanded, so D is not added again.

Question 4: Search with Partial Information – Belief States

A belief state is the set of physical states that remain possible given the agent's information. L = Left, R = Right, D = Dirty, and C = Clean.

State	Physical state (Agent, Left, Right)	State	Physical state (Agent, Left, Right)
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

(a) Problem Formulation

Component	Description
Initial belief state	{S1, S2, S3, S4, S5, S6, S7, S8}
Actions	Left, Right, and Suck
Goal test	Every physical state in the belief state has both squares clean.
Path cost	Each action costs 1; total cost is the number of actions.

(b) Sensorless action sequence: [Right, Suck, Left, Suck]

Step	Action	Belief state after action
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
3	Left	{S3, S7}
4	Suck	{S7}

The final belief state is {S7}, representing (L, C, C). Both squares are clean.

(c) Belief-State Prediction and Percept Update

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}

1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(d) Number of Possible Belief States

With 8 physical states, the number of possible subsets is 2■ = **256 belief states**, including the empty set. This can make search harder because the number of possible belief states grows exponentially with the number of physical states.

(e) Interpretation

Percepts shrink the belief state by removing physical states that do not match the observation. In the sensorless case, the agent only predicts the results of actions and cannot use observations to eliminate possibilities. The sensorless plan is still rational because [Right, Suck, Left, Suck] cleans both squares from every possible initial state. With each action costing 1, its plan cost is 4.